

# FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

## SSC PART-I (CLASS 9) PRACTICE EXAMINATION

### CHEMISTRY PAPER 25 | ANSWER KEY (Version A)

Syllabus: Chapter 10, Acids, Bases and Salts (35 marks) + Revision: Chapters 6 and 8 (15 marks)

Total Marks: 50

Chapter 10: 35

Revision: 15

**Marking rule used on this paper:** a missing **name** or a missing **unit** loses  $\frac{1}{2}$  **mark**, the same in every question. It never cancels a correct equation or correct working.

#### SECTION A: ANSWER KEY ( $10 \times 1 = 10$ MARKS)

**1. Cannot be classified as an Arrhenius acid: (c)  $\text{CO}_2$  1**

An Arrhenius acid must ionize in water to give  $\text{H}^+$ .  $\text{CO}_2$  has **no hydrogen**, so it cannot, even though it is acidic. This is one of the limitations of the Arrhenius theory.

Source: Ch 10, Review Question 1(i), book p147, and section 10.1.2, p142.

**2. A Bronsted-Lowry base: (a)  $\text{NH}_3$  1**

$\text{NH}_3$  accepts a proton to become  $\text{NH}_4^+$ .  $\text{HCl}$  and  $\text{CH}_3\text{COOH}$  are proton donors (acids), and  $\text{H}_3\text{O}^+$  is itself an acid.

Source: Ch 10, Review Question 1(ii), book p147.

**3. Salt formed when  $\text{Mg}(\text{OH})_2$  neutralizes  $\text{HCl}$ : (c)  $\text{MgCl}_2$  1**

$\text{Mg}(\text{OH})_2 + 2\text{HCl} \rightarrow \text{MgCl}_2 + 2\text{H}_2\text{O}$ . The salt takes the metal from the base and the Cl from hydrochloric acid.

Source: Ch 10, Review Question 1(iii), book p147; neutralization, section 10.3.

**4. Ammonia is a Bronsted-Lowry base because it: (d) accepts a proton 1**

Ammonia contains **no OH group**, which is exactly why the Arrhenius theory could not explain it. Bronsted-Lowry does: it accepts a proton and becomes  $\text{NH}_4^+$ .

Source: Ch 10, Review Question 1(iv), book p147, and section 10.1.2, p142.

**5. A weak base: (b)  $\text{Al}(\text{OH})_3$  1**

The book: "A base that ionizes to a little extent is called a weak base...  $\text{Al}(\text{OH})_3$ ,  $\text{NH}_3$  etc are weak bases."  $\text{NaOH}$ ,  $\text{KOH}$  and  $\text{Ca}(\text{OH})_2$  are its examples of strong bases.

Source: Ch 10, section 10.2.2, book p144.

**6. A property acids and bases share: (d) their aqueous solutions conduct electricity 1**

In Table 10.1 the two shared rows are **corrosive to skin** and **aqueous solutions conduct electricity**. Only the second is offered here.

Source: Ch 10, Table 10.1, book p141.

**7. A strong base: (a) completely dissociates in aqueous solution 1**

Strength means **how completely** it ionizes, not how well it dissolves. A weak base dissociates only partially.

Source: Ch 10, Key Points and section 10.2.2, book p144 and p147.

**8. [REVISION, Ch 8]  $\Delta H^\circ = +131.4 \text{ kJ}$  means the reaction is: (b) endothermic 1**

$\Delta H$  is **positive for endothermic** reactions (heat is absorbed) and negative for exothermic ones. The book lists this exact reaction under endothermic reactions.

Source: Ch 8, sections 8.2.2 and 8.3, book p129.

**9. [REVISION, Ch 6] Mass of 0.75 mol of  $\text{CO}_2$ : (b) 33 g 1**

Molar mass =  $12 + 2(16) = 44 \text{ g/mol}$ . Mass =  $0.75 \times 44 = 33 \text{ g}$ .

**10. [REVISION, Ch 6] Moles in 54 g of water: (a) 3 mol** 1

Molar mass =  $2(1) + 16 = 18$  g/mol. Moles =  $54 \div 18 = 3$  mol. 0.33 mol is the trap: that comes from dividing 18 by 54.

**SECTION B: ANSWER KEY ( $6 \times 4 = 24$  MARKS)****2. Arrhenius acids, strong acid, weak base.** 4

(i) An Arrhenius acid is a substance that **ionizes in water to produce  $H^+$  ions**. 1 Any two examples: HCl,  $HNO_3$ ,  $H_2SO_4$ . 1

(ii) HCl **ionizes completely (100%)** in water; all of its molecules ionize. 1

(iii) Ammonia **ionizes only to a little extent** in water, so it produces fewer  $OH^-$  ions. 1

Source: Ch 10, sections 10.1.1 (p141), 10.2.1 and 10.2.2 (p144); Review Question 2(ii), (iii), (v), p148.

**3. Limitations of the Arrhenius theory; Bronsted-Lowry explanation.** 4

(i) Any two, 1 mark each:

- It applies **only to aqueous solutions**.
- It cannot explain why  $CO_2$  and  $SO_2$  are **acidic**: they contain no H.
- It cannot explain why  $NH_3$  is a **base**: it contains no OH.

(ii)  $HNO_3$  **donates a proton ( $H^+$ )**, so it is the acid. 1  $NH_3$  **accepts that proton** and becomes  $NH_4^+$ , so it is the base.

1

Source: Ch 10, section 10.1.2, book p142; Review Question 5(ii), p148.

**4. Bronsted-Lowry pair; ionization of two bases.** 4

(i)  $HNO_3$  **is the acid**: it donates a proton and becomes  $NO_3^-$ . 1  $H_2O$  **is the base**: it accepts the proton and becomes  $H_3O^+$ . 1  $H_3O^+$  is a product; naming it as the acid earns 0 for that part.

(ii)(a)  $KOH \rightarrow K^+ + OH^-$  (single arrow)  $\frac{1}{2}$

(b)  $NH_4OH \rightleftharpoons NH_4^+ + OH^-$  (reversible arrow)  $\frac{1}{2}$

Correct arrows (single for KOH, reversible for  $NH_4OH$ )  $\frac{1}{2}$ .  $NH_4OH$  is the weak base  $\frac{1}{2}$ .

Source: Ch 10, section 10.1.1 (p142, which prints both equations with these arrows) and 10.2.2 (p144); Review Question 5(i), p148.

**5. Neutralization equations.** 4

(i)  $NH_4OH + HNO_3 \rightarrow NH_4NO_3 + H_2O$  2

Already balanced as written: 1 mark for the correct formulas, 1 for the balanced equation.

(ii)  $H_2SO_4 + Mg(OH)_2 \rightarrow MgSO_4 + 2H_2O$  2

1 mark for the correct formulas, 1 for the 2 in front of  $H_2O$  (count H:  $2 + 2 = 4$  on the left).

Source: Ch 10, Review Questions 3 and 4(i), book p148.

**6. Table 10.1 comparison.** 4

| Property                   | Acid         | Base          |
|----------------------------|--------------|---------------|
| (i) Taste                  | Sour         | Bitter        |
| (ii) Effect on blue litmus | Turns it red | No effect     |
| (iii) Effect on red litmus | No effect    | Turns it blue |
| (iv) Effect on skin        | Corrosive    | Corrosive     |

$\frac{1}{2}$  mark for each of the eight cells.

Source: Ch 10, Table 10.1, book p141. Note row (iv): both are corrosive in the book's table.

7. [REVISION, Ch 8]  $\Delta H$  for  $2\text{HI} \rightarrow \text{H}_2 + \text{I}_2$ . 4

$\Delta H$  = Sum of bond dissociation energies of reactants – Sum of bond formation energies of products 1

$$\Delta H = [2 \times 299] - [436 + 151] = 598 - 587 \quad 1$$

$\Delta H = +11 \text{ kJ/mol}$  1 (sign and unit required; a missing unit loses ½)

$\Delta H$  is **positive**, so the reaction is **endothermic**: more energy is absorbed breaking the H–I bonds than is released forming H–H and I–I. 1

Source: Ch 8, section 8.4, book p129–130. The data are the book's Example 8.1, which does the **forward** reaction and gets  $-11 \text{ kJ/mol}$ . This is the reverse on purpose: reversing the reaction reverses the sign of  $\Delta H$ . A student who copies  $-11$  from the example loses the answer and conclusion marks.

## SECTION C: ANSWER KEY (2 × 8 = 16 MARKS)

### 8. Proton donors and proton acceptors. 8

(i) 1 mark for each identification **with** its reason:

| Reaction                                                                                             | Bronsted-Lowry acid                                                | Bronsted-Lowry base                                                          |
|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|------------------------------------------------------------------------------|
| (a) $\text{HCO}_3^- + \text{H}_2\text{O} \rightleftharpoons \text{CO}_3^{2-} + \text{H}_3\text{O}^+$ | $\text{HCO}_3^-$ : donates a proton and becomes $\text{CO}_3^{2-}$ | $\text{H}_2\text{O}$ : accepts the proton and becomes $\text{H}_3\text{O}^+$ |
| (b) $\text{HCl} + \text{HCO}_3^- \rightleftharpoons \text{H}_2\text{CO}_3 + \text{Cl}^-$             | $\text{HCl}$ : donates a proton and becomes $\text{Cl}^-$          | $\text{HCO}_3^-$ : accepts the proton and becomes $\text{H}_2\text{CO}_3$    |

(ii)  $\text{HCO}_3^-$  donates a proton in (a), acting as an acid, and accepts a proton in (b), acting as a base. 1 A substance that can act as both an acid and a base is called **amphoteric**, just like water. 1

(iii) Water is a **proton acceptor** 1, because it takes the proton from  $\text{HS}^-$  and becomes  $\text{H}_3\text{O}^+$ . 1

Source: Ch 10, Review Question 7(ii), (iv) and (v), book p148; amphoteric substances, section 10.1.2, p143. The acid is always on the **left** of the arrow:  $\text{H}_3\text{O}^+$ ,  $\text{H}_2\text{CO}_3$  and  $\text{CO}_3^{2-}$  are products.

### 9. [REVISION, Ch 6] Zinc and hydrochloric acid, 6.5 g of Zn. 8

(i) moles of Zn = mass ÷ molar mass =  $6.5 \div 65 = 0.1 \text{ mol}$  1

(ii) From the equation, 1 mol Zn needs 2 mol HCl, so moles of HCl =  $2 \times 0.1 = 0.2 \text{ mol}$  1; molar mass of HCl =  $1 + 35.5 = 36.5 \text{ g/mol}$ , so mass =  $0.2 \times 36.5 = 7.3 \text{ g}$  1

(iii) 1 mol Zn gives 1 mol  $\text{H}_2$ , so moles of  $\text{H}_2$  =  $0.1 \text{ mol}$  1; mass =  $0.1 \times 2 = 0.2 \text{ g}$  1

(iv) molecules of  $\text{H}_2$  =  $0.1 \times 6.02 \times 10^{23} = 6.02 \times 10^{22} \text{ molecules}$  1

(v) molar mass of  $\text{ZnCl}_2$  =  $65 + 2(35.5) = 136 \text{ g/mol}$ ; mass =  $0.1 \times 136 = 13.6 \text{ g}$  1

(vi) “6.5 g of zinc produces 0.2 g of hydrogen gas and 13.6 g of zinc chloride.” 1

Marking note: the 2 : 1 ratio in (ii) is the step to watch; using 0.1 mol of HCl loses that mark but carries forward correctly. Each answer needs its unit (½ off if missing). Reaction from Ch 10, section 10.3 (p145); mole calculations from Ch 6.

### Mark distribution check

| Section      | Chapter 10      | Revision                  | Total     |
|--------------|-----------------|---------------------------|-----------|
| A: MCQs      | 7 (Q1 to Q7)    | 3 (Q8 Ch 8; Q9, Q10 Ch 6) | 10        |
| B: Short     | 20 (Q2 to Q6)   | 4 (Q7, Ch 8)              | 24        |
| C: Long      | 8 (Q8)          | 8 (Q9, Ch 6)              | 16        |
| <b>Total</b> | <b>35 (70%)</b> | <b>15 (30%)</b>           | <b>50</b> |

Answer letter balance for Section A: a = 3 (Q2, Q7, Q10), b = 3 (Q5, Q8, Q9), c = 2 (Q1, Q3), d = 2 (Q4, Q6).